

THE HYPOTHALAMUS

1. General Function of the Hypothalamus

The **hypothalamus** is the major connection between the **nervous system and endocrine system**.

It coordinates the activity of the **pituitary gland** by producing chemical messengers, especially:

- **Releasing hormones**
- Other **peptides**
- **Amines**

These substances control pituitary hormone secretion.

The hypothalamus is also an important control center for many functions of the **autonomic nervous system**.

2. Location and Structure of the Hypothalamus

The hypothalamus is part of the **diencephalon**.

It forms the **floor of the third ventricle**.

Important structures associated with the hypothalamus include:

- **Optic chiasma**
- **Tuber cinereum**
- **Mammillary bodies**
- **Median eminence**

The following structures are sometimes not included as part of the hypothalamus itself, although they are actually extensions of it into the pituitary:

- **Infundibulum**
- **Neurohypophysis**

The neurohypophysis includes the **stalk of the posterior pituitary and the posterior lobe**.

3. What Does the Hypothalamus Control?

The hypothalamus produces peptides and amines that influence the pituitary gland.

These signals can cause the pituitary to produce:

A. Tropic hormones

These hormones act on another endocrine gland and stimulate it to produce hormones.

Example:

Hypothalamus → pituitary ACTH → adrenal cortex → cortisol

ACTH is therefore a **tropic hormone** because it stimulates another endocrine gland.

B. Hormones that directly produce a biological effect

Some pituitary hormones act more directly on tissues.

Example:

Prolactin (PRL) acts directly on tissues involved in milk production.

THE PITUITARY GLAND (HYPOPHYSIS)

4. Parts of the Pituitary Gland

The pituitary gland, also called the **hypophysis**, consists of four major parts:

1. **Adenohypophysis** – anterior pituitary
 - Mainly the **pars distalis**
 2. **Neurohypophysis** – posterior pituitary
 - Mainly the **pars nervosa**
 3. **Pars intermedia** – intermediate lobe
 4. **Pars tuberalis**
-

5. Embryonic Origin of the Pituitary

The anterior and posterior pituitary have **different embryonic origins**.

Adenohypophysis

- Develops from an upward extension of the embryonic **oral ectoderm**.
- This structure is called **Rathke's pouch**.

Neurohypophysis

- Develops from a downward extension of **neural ectoderm** from the floor of the third ventricle.

The two developing structures grow toward each other and form the pituitary gland.

Easy reminder:

Anterior pituitary → Rathke's pouch → oral ectoderm

Posterior pituitary → neural ectoderm → hypothalamus

THE NEUROHYPOPHYSIS / POSTERIOR PITUITARY

6. The Posterior Pituitary Is an Extension of the Hypothalamus

The neurohypophysis is different from the anterior pituitary because its **nerve cell bodies are actually located in the hypothalamus**.

Most of these cell bodies are located in:

- **Supraoptic nucleus**
- **Paraventricular nucleus**

Their axons extend from the hypothalamus down into the posterior pituitary.

Structure:

Hypothalamus (cell bodies) → axons → pituitary stalk → posterior pituitary (nerve endings)

Therefore:

Posterior pituitary = extension of the hypothalamus

7. Neurosecretory Neurons

The neurons that produce posterior pituitary hormones are called **neurosecretory neurons**.

They are different from ordinary neurons.

Neurosecretory neurons:

1. **Do not innervate other neurons**
 - Although they themselves receive neural input.
2. Release their secretory products into the **bloodstream**.
3. Their hormones can travel to and act on **distant tissues**.

This makes them a direct connection between the **nervous system and endocrine system**.

Another important difference:

Anterior pituitary hormones often stimulate other tissues or endocrine glands to produce hormones.

Posterior pituitary hormones can **directly cause the desired response in their target tissues**.

8. Main Posterior Pituitary Hormones

The two major hormones associated with the neurohypophysis are:

- **Oxytocin**
- **Vasopressin**, also called **antidiuretic hormone (ADH)**

Important point:

Although they are released from the posterior pituitary, they are actually **synthesized in the hypothalamus**.

9. Synthesis and Transport of Oxytocin and Vasopressin

Both hormones are produced by **magnocellular neurons**.

"Magnocellular" means the neurons have **large cell bodies**.

Both **oxytocin and vasopressin are produced in both the supraoptic and paraventricular nuclei**.

They are not strictly produced by only one nucleus.

10. Formation of Vasopressin

Vasopressin is first synthesized in the hypothalamic cell body as a **preprohormone** called:

Prepropressophysin

The process is:

Preprohormone → removal of "pre" portion → prohormone → transported down axon → further cleavage → vasopressin

A peptide fragment remaining after processing is called **neurophysin II**.

11. Formation of Oxytocin

Oxytocin is initially produced as the preprohormone:

Preprooxyphysin

The "pre" portion is removed, and the resulting prohormone is packaged into granules.

As the granules move down the axon, the prohormone is cleaved to produce:

Oxytocin + neurophysin I

The physiological function of the neurophysins is **not currently known**.

Additional point:

Neurophysin I is released into the blood along with oxytocin and can be measured as an alternative indicator of **oxytocin release**.

12. Transport to the Posterior Pituitary

The hormones are made in the **cell bodies in the hypothalamus**.

The secretory granules are then transported down the axons to the **posterior pituitary**.

There, the hormones are stored in the **nerve terminals** until a stimulus causes their release.

Easy flow:

Hypothalamic cell body → hormone synthesis → secretory granules → axonal transport → posterior pituitary → storage in nerve endings

13. Release of Oxytocin and Vasopressin

Hormone release begins when the hypothalamic neuron is stimulated by **neural input**.

Sequence:

Neural stimulation

→ depolarization of neuron

→ action potential travels down the axon

→ reaches nerve terminal

→ Ca^{2+} enters the nerve terminal

→ secretory granules fuse with the membrane

→ **exocytosis**

→ oxytocin or vasopressin enters the bloodstream

Key requirement:

Calcium (Ca^{2+}) is required for hormone release.

OXYTOCIN

14. Main Effects of Oxytocin

Oxytocin mainly causes **smooth muscle contraction**.

Its major targets include:

Mammary gland

It causes contraction of **myoepithelial cells** surrounding the mammary alveoli.

This pushes milk out of the alveoli and into the ducts and cisterns.

Suckling → hypothalamus → oxytocin → myoepithelial contraction → milk ejection Uterus

Oxytocin causes contraction of the **myometrium** of the uterus.

Therefore, the major actions of oxytocin are:

Mammary gland → milk ejection

Uterus → smooth muscle contraction

VASOPRESSIN / ADH

15. Main Functions of Vasopressin

Vasopressin has two major actions:

Primary effect: Antidiuresis

Its most important function is to help the kidneys **retain water**.

This is why vasopressin is also called:

Antidiuretic hormone (ADH)

It is the **most important hormone for controlling body water balance**.

Secondary effect: Vasopressor action

Vasopressin can also cause **contraction of vascular smooth muscle**.

This increases **blood pressure**.

Easy reminder:

ADH → saves water

Vasopressin → also raises blood pressure

16. Forms of Vasopressin in Different Species

The major form differs among species:

Species	Main hormone form
Most species	Arginine vasopressin (AVP)
Pigs	Lysine vasopressin
Birds	Arginine vasotocin

CONTROL OF VASOPRESSIN SECRETION

17. Plasma Osmolality Controls ADH Release

The most important stimulus for vasopressin release is a change in **plasma osmolality**.

Osmolality = concentration of dissolved substances in body fluids relative to water.

When plasma osmolality increases:

This usually means there is **too little water relative to solutes**.

↑ Plasma osmolality

→ hypothalamic **osmoreceptors** increase their firing

→ stimulate vasopressin-producing neurons

→ **↑ vasopressin release**

→ kidneys retain more water

→ body water increases

→ osmolality moves back toward normal

This is a **negative-feedback system**.

The body normally keeps the solute-to-water ratio within approximately **1–2% of normal**.

18. Other Receptors Involved in Water Balance

The body also has receptors in the:

- **Esophagus**
- **Stomach**

These can detect **water intake very quickly**.

This allows the body to respond to drinking even before major changes in plasma osmolality occur.

19. Blood Volume Also Controls Vasopressin

Vasopressin secretion is also influenced by **blood volume**.

When blood volume decreases:

↓ Blood volume

→ stretch receptors in the **atria** detect the change

→ neural pathways alter inhibition of the osmoreceptor system

→ vasopressin release increases

→ kidney retains more water

→ blood volume is supported

Changes that reduce blood pressure also stimulate vasopressin release through:

- **Baroreceptors in the carotid sinus**
- **Baroreceptors in the aortic arch**

Important idea:

↑ Osmolality → ↑ ADH

↓ Blood volume → ↑ ADH

↓ Blood pressure → ↑ ADH

DIABETES INSIPIDUS (DI)

20. What Is Diabetes Insipidus?

Diabetes insipidus (DI) is a disorder in which the body cannot properly conserve water because **ADH is absent or ineffective**.

The major signs are:

- **Polyuria** = excessive urine production
- **Polydipsia** = excessive thirst and water intake
- **Dilute urine** with low specific gravity or osmolality

21. Types of Diabetes Insipidus

There are two major types:

A. Central diabetes insipidus

The problem is **insufficient ADH secretion**.

There is an absolute or relative deficiency of circulating ADH.

Central DI may be:

- **Primary**

- Idiopathic
- Congenital
- **Secondary**
 - Commonly associated with **head trauma**
 - **Neoplasia** can also cause it

B. Nephrogenic diabetes insipidus

ADH may be present, but the **kidneys do not respond properly to it**.

Simple comparison:

Central DI = not enough ADH

Nephrogenic DI = ADH present, but kidney cannot respond

22. Clinical Signs of Diabetes Insipidus in Dogs and Cats

Central DI can occur:

- At any age
- In any breed
- In either sex

However, **young adults around 6 months of age** are most commonly affected.

Major signs include:

- **Profound polyuria**
- **Polydipsia**
- **Nocturia** = urinating during the night
- **Urinary incontinence**
- Signs often present for **several months**

A commonly reported urine/water intake pattern is:

>100 mL/kg/day

compared with a normal range of approximately:

40–70 mL/kg/day

The severity depends on whether the ADH problem is **partial or complete**.

Other possible signs:

- **Weight loss**
- **Dehydration**

Animals may lose weight partly because they are constantly seeking water.

23. Laboratory Findings in DI

Routine tests are often surprisingly normal.

CBC and serum chemistry

Usually:

Normal

Plasma osmolality

Often **high** because of dehydration:

>310 mOsm/L

This can occur in both:

- Central DI
- Nephrogenic DI

Urinalysis

Usually unremarkable except for **persistently dilute urine**.

Typical urine specific gravity:

1.004–1.012

24. Primary Polydipsia vs DI

Primary polydipsia is excessive drinking that is **not caused by central or nephrogenic DI**.

Because these animals drink too much water, they may become overhydrated.

Therefore:

Primary polydipsia → low plasma osmolality

Often:

<290 mOsm/L

This contrasts with DI, where plasma osmolality is often increased because of dehydration.

25. Secondary Changes From Dehydration

If the animal is prevented from drinking enough water, dehydration may cause findings such as:

- Slightly increased **hematocrit**
- **Hypernatremia**

These findings are usually **secondary to dehydration**, rather than being the primary disorder.

26. Diagnosing Diabetes Insipidus

Tests used to distinguish:

- Central DI
- Nephrogenic DI
- Primary polydipsia

include:

A. Modified water deprivation test

This determines:

1. Whether the body releases **endogenous ADH** when dehydrated.
2. Whether the kidneys can **respond to ADH**.

B. ADH supplementation / response test

The animal's response to administered ADH can help distinguish whether the problem is:

- Lack of ADH, or
 - Failure of the kidneys to respond to ADH.
-

27. Important Safety Point About Water Deprivation Testing

Common causes of **polyuria and polydipsia (PU/PD)** should be ruled out before performing a water deprivation test.

This is especially important because:

Undiagnosed renal failure + water deprivation → potentially serious illness

Failure to recognize renal failure beforehand may result in:

- An incorrect diagnosis
- An inconclusive test
- Significant **morbidity**

SIADH

28. Syndrome of Inappropriate Antidiuretic Hormone Secretion

SIADH occurs when vasopressin is released **inappropriately**, even though there is no proper osmotic or blood-volume stimulus for its release.

In other words:

ADH is being secreted when the body does not actually need it.

Hypersecretion is **rare**.

29. Causes of SIADH

Neoplasia (tumors) is often associated with SIADH.

Some tumors produce ADH or ADH-like activity outside the normal hypothalamic-pituitary system. These are called **ectopic tumors**.

The **lungs** are a common location for the tumors associated with SIADH.

HIGH-YIELD SUMMARY

Hypothalamus

- Links the **nervous and endocrine systems**
- Produces releasing hormones and other peptides/amines
- Controls the pituitary
- Also controls many **autonomic nervous system** functions

Pituitary

Adenohypophysis = anterior pituitary

Neurohypophysis = posterior pituitary

Embryology:

Anterior → Rathke's pouch → oral ectoderm

Posterior → neural ectoderm → hypothalamus

POSTERIOR PITUITARY

Oxytocin

Produced in hypothalamus → transported down axons → released from posterior pituitary

Main effects:

- **Mammary gland → milk ejection**
- **Uterus → contraction**

Vasopressin/ADH

Main effects:

- **Kidney → water retention**
- **Blood vessels → vasoconstriction → ↑ blood pressure**

VASOPRESSIN CONTROL

↑ **Plasma osmolality** → ↑ **ADH** → ↑ **water retention**

↓ **Blood volume** → ↑ **ADH**

↓ **Blood pressure** → ↑ **ADH**

DIABETES INSIPIDUS

Central DI

↓ **ADH secretion**

Nephrogenic DI

Kidneys do not respond to ADH

Typical signs:

Polyuria + polydipsia + dilute urine

Typical DI findings:

- Plasma osmolality often **>310 mOsm/L**
- Urine specific gravity approximately **1.004–1.012**

Primary polydipsia:

- Plasma osmolality often **<290 mOsm/L**

Diagnosis:

Modified water deprivation test + ADH response

Rule out renal failure and other common causes of PU/PD first.

SIADH

Inappropriately high ADH secretion despite no appropriate osmotic or volume stimulus

Often associated with **neoplasia**, especially **ectopic tumors involving the lungs**.

THE ANTERIOR PITUITARY

1. Main Hormones of the Anterior Pituitary

The **adenohypophysis (anterior pituitary)** includes the:

- **Pars distalis**
- **Pars intermedia**

The six major hormones associated with the anterior pituitary are:

1. **Growth hormone (GH)** or **somatotropin**
2. **Prolactin (PRL)**
3. **Thyroid-stimulating hormone (TSH)** or **thyrotropin**

4. **Follicle-stimulating hormone (FSH)**
5. **Luteinizing hormone (LH)**
6. **Corticotropin (ACTH) or adrenocorticotrophic hormone**

2. Growth Hormone and Prolactin

GH and PRL are called **somatomammotropins**.

Growth hormone (GH)

- Produced by **somatotropes**
- Somatotropes are **acidophilic cells**
- GH is a **single-chain protein**
- Has **two disulfide bonds**

Prolactin (PRL)

- Produced by **lactotropes**
- PRL is also a **single-chain protein**
- Has **three disulfide bonds**

Relationship between GH and PRL

GH and PRL have some overlapping effects because about **50% of their amino acid sequences are similar**.

However:

GH is strongly species-specific in its biological activity.

This means GH from one species may not produce the same biological effects in another species.

3. TSH, FSH, and LH

TSH, FSH, and LH are produced by:

- **Thyrotropes → TSH**
- **Gonadotropes → FSH and LH**

These three hormones are **glycoproteins**, meaning they contain **carbohydrate groups** attached to the protein.

All three have:

- **α (alpha) subunit**
- **β (beta) subunit**

The two subunits are held together by **noncovalent bonds**.

Alpha subunit

- The **same among TSH, FSH, and LH**
- They are essentially **interchangeable**

Beta subunit

- **Different for each hormone**
- Determines the hormone's **specific biological action**

Easy rule:

α subunit = shared/common

β subunit = specific function

4. Other Glycoprotein Gonadotropins

Other hormones belong to the same general glycoprotein family, although they are not produced by the anterior pituitary.

Examples include:

- **Equine chorionic gonadotropin (eCG)**, formerly called pregnant mare's serum gonadotropin
- **Primate chorionic gonadotropin**

These are produced by **placental cells**.

ACTH AND THE PROOPIOMELANOCORTIN FAMILY

5. ACTH Comes From a Common Prohormone

ACTH and **β-lipotropin** belong to the **proopiomelanocortin (POMC) family**.

This means both are made from the same large precursor molecule:

POMC → cleavage → ACTH and other peptides

POMC is produced in both:

- **Pars distalis**
- **Pars intermedia**

However, the final products are different in each region.

6. Pars Distalis

In the **pars distalis**:

POMC → ACTH

ACTH is produced by **corticotropes**.

ACTH then acts mainly on the **adrenal cortex** to stimulate production of adrenal hormones, especially **cortisol**.

7. Pars Intermedia

In the **pars intermedia**:

ACTH → α-melanocyte-stimulating hormone (α-MSH)

α-MSH is the **predominant hormone of the pars intermedia**.

Another peptide fragment is also produced:

Corticotropin-like intermediate lobe peptide

Its physiological function is **not known**.

8. β-Lipotropin

In both the **pars distalis** and **pars intermedia**:

β-lipotropin → β-endorphins + γ-lipotropin

β-Endorphins

- Have **opioid activity**
 - May help regulate or **modulate gonadotropin secretion**
-

HYPOTHALAMIC CONTROL OF THE ANTERIOR PITUITARY

9. How the Hypothalamus Controls the Anterior Pituitary

For many years, scientists did not understand how the brain controlled the anterior pituitary.

The key discovery was the **hypothalamic-pituitary portal blood system**.

Important historical discovery

In the **1930s**, Popa and Fielding described the blood vessels connecting the hypothalamus and pituitary.

Around **1950**, Geoffrey Harris concluded that blood flows:

Hypothalamus → anterior pituitary

through a **portal blood system**.

10. Hypophyseal Portal System

The blood supply follows this general pathway:

Dorsal hypophyseal artery

→ capillary network in the **median eminence**

→ portal veins

→ capillary/sinusoidal network in the **pars distalis**

The **dorsal hypophyseal artery** supplies the **adenohypophysis**.

The **ventral hypophyseal artery** supplies the **neurohypophysis**.

The portal system has two major venous pathways supplying different areas of the pars distalis:

- One supplies the **ventral/central region**
- One supplies the **dorsal/peripheral region**

Main purpose of the portal system

It allows hypothalamic hormones to travel **directly to the anterior pituitary** without first being diluted throughout the general circulation.

11. Hypophysiotropic (Hypothalamic Releasing) Hormones

The hypothalamus produces regulatory hormones called **hypophysiotropic hormones**.

These are released into the **median eminence** and then enter the **portal blood system**.

They travel to the anterior pituitary and control the secretion of its hormones.

Basic pathway:

Hypothalamus → releasing/inhibiting hormone → portal blood → anterior pituitary → pituitary hormone

The production of these hypothalamic regulatory hormones is itself controlled by:

- **Neural signals**
- **Hormonal signals**

Some of these pituitary-related hormones are also found in other areas, including:

- **Brain**

- **Gastrointestinal tract**
- **Pancreas**

12. Major Hypothalamic Hormones

Hypothalamic hormone	Abbreviation	Main pituitary effect
Corticotropin-releasing hormone	CRH	↑ ACTH
Gonadotropin-releasing hormone	GnRH	↑ FSH and LH
Thyrotropin-releasing hormone	TRH	↑ TSH
Growth hormone-releasing hormone	GHRH	↑ GH
Somatostatin / GH-inhibiting hormone	GHIH	↓ GH
Dopamine / prolactin-inhibiting hormone	PIH	↓ PRL; also inhibits TSH
Prolactin-releasing factor	PRF	Stimulates PRL; source is uncertain

All of these hypothalamic hormones are peptides except dopamine, which is a catecholamine.

13. Important Details About Each Hypothalamic Hormone

CRH

- **41-amino-acid peptide**
- Stimulates corticotropes
- Causes release of **ACTH** and other **POMC-derived peptides**

GnRH

- **Decapeptide** = 10 amino acids
- Stimulates gonadotropes to release:
 - **FSH**
 - **LH**

TRH

- **Tripeptide** = 3 amino acids
- Stimulates thyrotropes to release **TSH**

GHRH

- **44-amino-acid polypeptide**
- Stimulates somatotropes to release **GH**

Somatostatin

- **Tetradecapeptide** = 14 amino acids
- Inhibits secretion of **GH**

Dopamine

- A **catecholamine**
- Inhibits:

- Prolactin secretion
- TSH secretion

TROPIC HORMONES

14. Which Anterior Pituitary Hormones Are Tropic?

A **tropic hormone** mainly acts on another endocrine tissue and stimulates it to produce hormones.

Originally, four anterior pituitary hormones were considered tropic:

- FSH
- LH
- TSH
- ACTH

GH is now also considered tropic

GH stimulates the **liver** to produce **somatomedins**, especially:

Insulin-like growth factor-1 (IGF-1)

IGF-1 then contributes to the **negative feedback regulation of GH**.

PRL

PRL remains the main anterior pituitary hormone for which **negative feedback from its target tissue hormones has not been demonstrated**.

FEEDBACK CONTROL OF THE ANTERIOR PITUITARY

15. Negative Feedback Is the Main Control Mechanism

The secretion of anterior pituitary protein hormones is mainly controlled by **negative feedback**.

The target organ's hormone feeds back to reduce further stimulation from the hypothalamus and/or pituitary.

16. Long-Loop Feedback

In a **long-loop feedback system**, the hormone produced by the target endocrine organ feeds back to both:

- Hypothalamus
- Anterior pituitary

Example: ACTH and cortisol

Hypothalamus → CRH → anterior pituitary → ACTH → adrenal cortex → cortisol

Then:

Cortisol → inhibits CRH + inhibits ACTH

This prevents excessive cortisol production.

Easy flow:

CRH → ACTH → cortisol

Cortisol –| CRH and ACTH

17. Short-Loop Feedback

A **short-loop feedback system** occurs when an **anterior pituitary hormone directly inhibits the hypothalamus**.

Example:

ACTH –| CRH

So:

Pituitary hormone → inhibits hypothalamic releasing hormone

18. Pulsatile Secretion of Gonadotropins

Negative feedback does not mean that pituitary hormones are released continuously at a constant rate.

For example, even though **estrogen continuously inhibits gonadotropin secretion**, FSH and LH are released in **pulses**.

The gonadotropins alternate between:

Secretion → no secretion → secretion → no secretion

The **ovarian hormonal environment** influences the:

- Pulse frequency
- Pulse amplitude

Progesterone-dominant state

- Lower pulse frequency
- Higher pulse amplitude

Estrogen-dominant state

- Higher pulse frequency
- Lower pulse amplitude

This demonstrates that hormone secretion is **dynamic and pulsatile**, rather than constant.

CLINICAL APPLICATION: GROWTH HORMONE DISORDERS

19. Growth Hormone Deficiency: Pituitary Dwarfism

Pituitary dwarfism occurs when GH production is insufficient because of **damage or abnormal development of the pituitary gland**.

Possible causes include:

- Neoplasia
- Degenerative disease
- Developmental abnormalities

Other pituitary hormones may also be decreased, including:

- TSH
- ACTH
- LH
- FSH
- GH

This condition may therefore result in **panhypopituitarism** when multiple pituitary hormones are deficient.

20. Pituitary Dwarfism in Dogs

It is most commonly seen in:

German Shepherd dogs, usually 2–6 months old

Other reported breeds include:

- Canelian bear dogs
- Spitz-type dogs
- Toy Pinschers
- Weimaraners

In German Shepherds, it is inherited as an:

Autosomal recessive trait

and is associated with a **cystic Rathke's pouch**.

21. Clinical Signs of Pituitary Dwarfism

The first signs usually appear during the first **2–3 months of life**.

Major signs:

- **Slow growth**
- **Proportionate dwarfism**
- **Mental retardation/developmental delay**
- Difficulty with **house training**

Physical findings:

- Retained **puppy hair coat**
 - **Hypotonic skin**
 - **Truncal alopecia** (hair loss over the body)
 - **Hyperpigmentation**
 - **Infantile genitalia**
 - **Delayed dental eruption**
-

22. Clinicopathologic Findings

Possible laboratory abnormalities include:

- **Eosinophilia**
- **Lymphocytosis**
- Mild **normocytic, normochromic anemia**
- **Hypophosphatemia**
- Sometimes **hypoglycemia**

Hypoglycemia may occur because of **secondary adrenal insufficiency** when ACTH is also deficient.

23. Differential Diagnoses for Stunted Growth

Other causes of poor growth must be considered, including:

- **Hypothyroid dwarfism**

- **Portosystemic shunt**
 - **Diabetes mellitus**
 - **Hyperadrenocorticism**
 - **Malnutrition**
 - **Parasitism**
-

24. Diagnosis of Pituitary Dwarfism

Diagnosis may involve measuring:

- **Serum GH**
- **IGF-1 (somatomedin C)**

GH assays for dogs and cats are no longer commercially available in many settings.

Advantage of IGF-1

IGF-1 is:

- **More stable**
- **Not species-specific**

Therefore, it is useful for detecting GH-related disorders.

Affected dogs may also show **subnormal responses to TSH and ACTH stimulation tests**.

Endogenous:

- **TSH ↓**
- **ACTH ↓**

This occurs because multiple anterior pituitary hormones may be deficient.

ACROMEGALY / HYPERSOMATOTROPISM

25. What Is Acromegaly?

Acromegaly is caused by **chronic excessive GH secretion in an adult animal**.

Unlike pituitary dwarfism:

Pituitary dwarfism = too little GH

Acromegaly = too much GH

26. Causes of Acromegaly

Dogs

Canine acromegaly is **very rare**.

It is usually associated with administration of **progestational drugs** to intact female dogs to suppress estrus.

These hormones stimulate **mammary tissue to produce excessive GH**.

Cats

Feline acromegaly is usually caused by a:

GH-secreting anterior pituitary tumor

These tumors usually:

- Grow **slowly**
- May be present for a long time before obvious signs develop

It is most common in:

- **Older cats**
- Approximately **8–14 years old**
- More common in **males**

27. Common Clinical Signs of Acromegaly

The most common initial signs are often those of **poorly controlled diabetes mellitus**:

- **Polydipsia**
- **Polyuria**
- **Polyphagia**

A very important clue is:

Weight gain despite uncontrolled diabetes

Animals may develop an increase in **lean body mass**, rather than the expected weight loss associated with uncontrolled diabetes.

28. Organ Enlargement

Acromegaly can cause **organomegaly**.

Possible findings include:

- **Renomegaly**
- **Hepatomegaly**
- Enlargement of endocrine organs

29. Physical Changes

Some affected dogs and cats may develop enlargement of:

- **Paws/extremities**
- **Jaw**
- **Tongue**
- **Forehead**
- **Skull**

There may also be:

- Increased **muscle mass**
- Growth of the **acral portions** of the body

Acral areas include the:

- Paws
- Chin
- Skull

30. Cardiovascular and Renal Changes

These usually develop later in the disease.

Cardiovascular

- **Cardiomegaly**
- **Systolic murmurs**
- **Congestive heart failure**

Renal

Approximately **50% of acromegalic cats** may eventually develop **azotemia**.

31. Neurologic Signs

Neurological signs commonly described in human acromegaly are generally **not seen in small animals**.

Examples include:

- Peripheral neuropathy
- Paresthesias
- Carpal tunnel syndrome
- Sensory and motor defects
- Headache
- Visual field abnormalities

These are generally not typical findings in acromegalic dogs and cats.

32. Acromegaly and Diabetes Mellitus

Insulin resistance and impaired glucose tolerance occur in essentially all cats and most dogs with acromegaly.

This leads to:

↑ Insulin resistance → ↑ blood glucose → diabetes mellitus

Despite this severe insulin resistance:

- Serum **insulin concentrations are very high**
- **Ketosis is uncommon**

Important clinical clue

Suspect **acromegaly in a diabetic cat, especially a male cat, when the diabetes is extremely insulin-resistant**.

33. Diagnosis of Acromegaly

Definitive diagnosis requires evidence of increased:

- **GH**, or
- **IGF-1**

Because canine and feline GH assays are no longer commercially available in many settings, **IGF-1 is commonly used**.

IGF-1 is useful because it is:

- **Non-species-specific**
- More practical for testing

Imaging

In cats, the most definitive approach combines:

↑ Serum IGF-1 + CT or MRI of the pituitary region

The diagnosis should also consider and exclude other causes of insulin resistance, such as:

- **Hyperthyroidism**
- **Hyperadrenocorticism**

34. Important Frequency in Diabetic Cats

Acromegaly should be strongly considered in diabetic cats that fail to achieve remission despite appropriate treatment.

The notes report that approximately **30% of diabetic cats that do not go into remission with an ultra-low-carbohydrate diet and insulin therapy may be acromegalic.**

HIGH-YIELD SUMMARY

Anterior Pituitary Hormones

GH → growth; stimulates liver to produce **IGF-1**

PRL → reproductive/mammary functions

TSH → stimulates thyroid

FSH → gonadal function

LH → gonadal function/ovulation

ACTH → stimulates adrenal cortex

Cell Types

Cell	Hormone
Somatotrope	GH
Lactotrope	PRL
Thyrotrope	TSH
Gonadotrope	FSH + LH
Corticotrope	ACTH

Hypothalamus → Anterior Pituitary

Hypothalamic hormone	Effect
CRH	↑ ACTH
GnRH	↑ FSH + LH
TRH	↑ TSH
GHRH	↑ GH
Somatostatin	↓ GH
Dopamine	↓ PRL and ↓ TSH

Feedback

Long-loop:

Target gland hormone → inhibits hypothalamus + pituitary

Example:

CRH → **ACTH** → **cortisol**

Cortisol –| **CRH + ACTH**

Short-loop:

Pituitary hormone → inhibits hypothalamus

Example:

ACTH –| **CRH**

Pituitary Disorders

Pituitary dwarfism

↓ GH

Usually:

- Young dogs
- Especially German Shepherds
- Slow growth
- Proportionate dwarfism
- Retained puppy coat
- Alopecia
- Hyperpigmentation
- Delayed dental eruption
- Possible multiple pituitary hormone deficiencies

Acromegaly

↑ GH in adults

Dogs:

Usually associated with progestational drugs

Cats:

Usually GH-secreting pituitary tumor

Common clues:

- **Insulin-resistant diabetes mellitus**
- Polyuria
- Polydipsia
- Polyphagia
- Increased lean body mass
- Organ enlargement
- Enlarged paws/jaw/tongue/skull

Best practical marker for both GH-related disorders:

IGF-1